Comprehensive Electronics Study Guide

BJT, MOSFET, and Amplifier Frequency Response

b4dnw

“Abandon all hope ye who enter here”

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1 Introduction

This study guide is designed to provide a comprehensive, student-friendly review of Bipolar Junction Transistors (BJT), Metal-Oxide-Semiconductor Field-Effect Transistors (MOSFET), and Amplifier Frequency Response. It focuses on conceptual understanding, mathematical derivations, and repeatable problem-solving procedures to prepare you for examinations.

Note: As specific lecture files were not attached, standard academic notation (e.g., Sedra/Smith conventions) is used throughout.

2 Bipolar Junction Transistors (BJT)

2.1 Learning Objectives

- Understand the physical structure and operation of NPN and PNP transistors.
- Master the mathematical relationships between terminal currents.
- Perform DC circuit analysis using systematic procedures.

2.2 Conceptual Explanation and Physical Operation

A BJT is a three-terminal semiconductor device consisting of three doped regions: Emitter (E), Base (B), and Collector (C). It operates as a current-controlled current source.

Physical Operation (Active Mode - NPN):

1. **Forward Bias (BE Junction):** A voltage applied across the Base-Emitter junction causes electrons to be injected from the heavily doped Emitter into the Base.
2. **Recombination in Base:** The Base is very thin and lightly doped. Only a small fraction of injected electrons recombine with holes here, forming the small base current (I_B).
3. **Reverse Bias (BC Junction):** The remaining electrons diffuse across the base and are swept into the Collector by the strong electric field of the reverse-biased Base-Collector junction, forming the large collector current (I_C).

2.3 Important Formulas

BJT Current Relationships (Active Region)

$$I_C = \beta I_B \tag{1}$$

$$I_E = I_B + I_C = (\beta + 1) I_B \tag{2}$$

$$\alpha = \frac{\beta}{\beta + 1} \implies I_C = \alpha I_E \tag{3}$$

Definitions:

- I_B, I_C, I_E: Base, Collector, and Emitter DC currents (A).
- β: Common-emitter current gain (typically 50–300).
- α: Common-base current gain (typically 0.98–0.99).

Assumptions/Validity: Valid ONLY in the Forward-Active region.

2.4 Circuit Analysis Procedure: BJT DC Analysis

Follow these repeatable steps during an exam to solve BJT DC biasing circuits:

1. **Identify Configuration:** Identify the bias type (e.g., Voltage Divider Bias, Base Bias).
2. **Assume Operating Region:** Always assume the transistor is in the **Forward-Active** region initially.
3. **Base-Emitter KVL:** Write Kirchhoff's Voltage Law (KVL) around the Base-Emitter loop. Assume $V_{BE} \approx 0.7$ V for NPN.
4. **Find Base Current:** Solve the KVL equation for I_B.
5. **Apply Relationships:** Calculate $I_C = \beta I_B$ and $I_E = (\beta + 1)I_B$.
6. **Collector-Emitter KVL:** Write KVL around the Collector-Emitter loop to find V_{CE}.
7. **Verify Assumption (Sanity Check):** Ensure $V_{CE} > V_{CE(sat)}$ (usually 0.2 V). If $V_{CE} < 0.2$ V, the assumption was wrong; the BJT is in Saturation. Recalculate using $V_{CE(sat)} = 0.2$ V.

2.5 Common Mistakes

- **Blindly using β:** Students often calculate $I_C = \beta I_B$ even when the transistor is in saturation. β is only valid in the active region! In saturation, $I_C < \beta I_B$ (forced beta).
- **PNP polarities:** Forgetting that for PNP, currents flow out of the collector/base and into the emitter. KVLs use V_{EB} (not V_{BE}).

2.6 Quick Revision Summary: BJT

- **Key Concept:** BJT is a current-controlled device. I_B controls I_C.
- **Active Mode:** BE forward-biased (0.7 V), BC reverse-biased. $I_C = \beta I_B$.
- **Saturation Mode:** Both junctions forward-biased. $V_{CE} = 0.2$ V.

3 MOS Field-Effect Transistors (MOSFET)

3.1 Conceptual Explanation and Physical Operation

A MOSFET is a four-terminal device (Gate, Drain, Source, Body). Unlike the BJT, it is a **voltage-controlled current source**.

Physical Operation (Enhancement NMOS):

1. **Subthreshold (Cutoff):** If Gate-to-Source voltage (V_{GS}) is less than the threshold voltage (V_t), no channel exists. Current $I_D = 0$.
2. **Channel Formation:** When $V_{GS} > V_t$, an electric field attracts electrons to the surface, creating an N-type inversion layer (channel) connecting Drain and Source.
3. **Triode Region:** If V_{DS} is small, the channel acts like a variable resistor controlled by V_{GS}.
4. **Saturation Region:** As V_{DS} increases and reaches $V_{DS(sat)} = V_{GS} - V_t$, the channel "pinches off" near the drain. The current I_D becomes relatively constant and independent of V_{DS} (ignoring channel-length modulation).

3.2 Important Formulas

NMOS Current Equations

1. Cutoff Region ($V_{GS} < V_t$):

$$I_D = 0 \tag{4}$$

2. Triode Region ($V_{GS} > V_t$ and $V_{DS} < V_{GS} - V_t$):

$$I_D = \mu_n C_{ox} \frac{W}{L} \left[(V_{GS} - V_t) V_{DS} - \frac{1}{2} V_{DS}^2 \right] \tag{5}$$

3. Saturation Region ($V_{GS} > V_t$ and $V_{DS} \geq V_{GS} - V_t$):

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_t)^2 (1 + \lambda V_{DS}) \tag{6}$$

Definitions:

- $\mu_n C_{ox}$: Process transconductance parameter (k'_n).
- W/L: Aspect ratio (Width/Length).
- V_t: Threshold voltage.
- λ: Channel-length modulation parameter (if $\lambda = 0$, ignore the $(1 + \lambda V_{DS})$ term).

3.3 Circuit Analysis Procedure: MOSFET DC Analysis

1. **Assume Operating Region:** Usually, assume the **Saturation Region** for amplifier circuits.
2. **Gate Current:** Remember that $I_G = 0$ A (ideal DC). Use this to find voltages at the gate easily (e.g., using a voltage divider).
3. **Gate-Source Loop KVL:** Express V_{GS} in terms of I_D if there is a source resistor.
4. **Apply MOSFET Equation:** Substitute V_{GS} into the Saturation equation: $I_D = \frac{1}{2} k_n (V_{GS} - V_t)^2$.

5. **Solve Quadratic:** Solve the resulting quadratic equation for I_D (or V_{GS}). *Crucial step: Discard the mathematical root that results in* $V_{GS} < V_t$.

6. **Drain-Source Loop KVL:** Calculate V_{DS} using KVL.

7. **Verify Assumption:** Check if $V_{DS} \geq V_{GS} - V_t$. If true, assumption holds. If false, re-solve using the Triode equation.

4 Amplifier Frequency Response

4.1 Conceptual Explanation

Real amplifiers do not amplify all frequencies equally. The **Frequency Response** describes how the amplifier's voltage gain (A_v) changes with input signal frequency.

- **Low Frequencies:** Gain drops due to external **coupling and bypass capacitors**. They act as open circuits at DC and high impedances at low frequencies.
- **Midband:** The flat region where gain is maximum and constant. Coupling caps act as short circuits; internal device caps act as open circuits.
- **High Frequencies:** Gain drops due to **internal parasitic capacitances** (e.g., C_{gs}, C_{gd} in MOSFETs) acting as short circuits and shunting the signal to ground.

4.2 Derivation: Miller's Theorem

Miller's theorem is essential for high-frequency analysis. It simplifies a circuit by converting a bridging impedance between two nodes into equivalent impedances to ground.
Setup: An impedance Z connects an input node (Node 1, voltage V_1) to an output node (Node 2, voltage V_2). The amplifier gain is $A = V_2/V_1$.
Derivation:

1. The current leaving Node 1 through Z is: $I_1 = \frac{V_1 - V_2}{Z}$
2. Substitute $V_2 = AV_1$: $I_1 = \frac{V_1 - AV_1}{Z} = \frac{V_1(1-A)}{Z}$
3. The equivalent impedance looking from Node 1 to ground is $Z_1 = \frac{V_1}{I_1}$:

$$Z_1 = \frac{Z}{1 - A} \tag{7}$$

4. If Z is a capacitor C, $Z = \frac{1}{j\omega C}$. Then $Z_1 = \frac{1}{j\omega C(1-A)}$.
5. This corresponds to an equivalent input **Miller Capacitance**:

Miller Capacitance Equation

$$C_{M1} = C(1 - A) \tag{8}$$

Physical Meaning: A small bridging capacitor (like C_{gd}) is magnified at the input by the amplifier's gain $(1 - A)$. Since A is usually a large negative number for inverting amplifiers (like Common Emitter), C_{M1} becomes very large, heavily restricting high-frequency bandwidth.

4.3 Procedure: Estimating High-Frequency Cutoff (f_H)

Use the **Open-Circuit Time Constant (OCTC)** method:

1. Identify all high-frequency internal capacitors (e.g., C_π, C_μ).
2. For each capacitor C_i, set the input source to zero (short voltage sources).
3. Assume all other high-frequency capacitors are **open circuits**.

4. Calculate the equivalent resistance $R_{eq,i}$ seen by capacitor C_i.
5. Compute the time constant $\tau_i = R_{eq,i} \times C_i$.
6. Sum the time constants: $\tau_H = \sum \tau_i$.
7. Calculate the upper cutoff frequency: $f_H \approx \frac{1}{2\pi\tau_H}$.

5 Supplementary Practice Questions

Q1 (Basic Conceptual - BJT): Why does the BJT collector current remain relatively constant in the active region despite increases in V_{CE} (ignoring Early effect)?

Q2 (Numerical - MOSFET DC): An NMOS transistor has $V_t = 1\,\text{V}$, $k_n = \mu_n C_{ox}\frac{W}{L} = 1\,\text{mA/V}^2$, and $\lambda = 0$. The gate is connected to $5\,\text{V}$, the source to ground, and the drain is connected to a $10\,\text{V}$ supply through a $2\,\text{k}\Omega$ resistor. Determine I_D and the operating region.

Q3 (Analysis - Frequency Response): An inverting amplifier has a midband gain of $A = -49\,\text{V/V}$. A feedback capacitor $C_f = 2\,\text{pF}$ is connected between input and output. Calculate the equivalent Miller capacitance at the input node.

6 Solutions to Practice Questions

Solution 1: In the active region, the Base-Collector junction is reverse-biased. The strong electric field sweeps any electrons that reach the depletion region across to the collector. The number of electrons arriving is determined solely by the Base-Emitter forward bias (which controls injection). Increasing V_{CE} widens the BC depletion region slightly (Early effect), but does not significantly change the number of carriers injected by the emitter.

Solution 2:

1. **Identify values:** $V_{GS} = 5\ \text{V} - 0\ \text{V} = 5\ \text{V}$.
2. **Assume Saturation:** $I_D = \frac{1}{2}k_n(V_{GS} - V_t)^2$.
3. **Calculate I_D:**
$$I_D = \frac{1}{2}(1\ \text{mA/V}^2)(5\ \text{V} - 1\ \text{V})^2 = \frac{1}{2}(1)(16) = 8\ \text{mA}.$$
4. **Calculate V_{DS} using KVL:**
$$V_{DS} = V_{DD} - I_D R_D = 10\ \text{V} - (8\ \text{mA})(2\ \text{k}\Omega) = 10\ \text{V} - 16\ \text{V} = -6\ \text{V}.$$
5. **Verify Assumption:** We need $V_{DS} \geq V_{GS} - V_t$. Here, $-6\ \text{V} \not\geq 4\ \text{V}$. The assumption is **WRONG**. The transistor is in the **Triode Region**.
6. **Recalculate (Triode):**
$$I_D = \frac{10 - V_{DS}}{2\text{k}}$$
$$I_D = k_n\left[(V_{GS} - V_t)V_{DS} - 0.5V_{DS}^2\right]$$
$$\frac{10 - V_{DS}}{2} = 1 \cdot \left[4V_{DS} - 0.5V_{DS}^2\right]$$
$$10 - V_{DS} = 8V_{DS} - V_{DS}^2 \implies V_{DS}^2 - 9V_{DS} + 10 = 0$$
Solving the quadratic yields $V_{DS} \approx 1.28\ \text{V}$ (valid, $< 4\ \text{V}$) or $7.71\ \text{V}$ (invalid).
7. **Final Answer:** $V_{DS} = 1.28\ \text{V}$, $I_D = \frac{10-1.28}{2\text{k}} = 4.36\ \text{mA}$. **Region: Triode.**

Solution 3:

1. **Identify variables:** $C = 2\ \text{pF}$, $A = -49$.
2. **Apply Miller's Equation:** $C_{M1} = C(1 - A)$.
3. **Calculate:**
$$C_{M1} = 2\ \text{pF} \times (1 - (-49)) = 2\ \text{pF} \times 50 = 100\ \text{pF}.$$
4. **Physical Meaning:** The amplifier's gain magnifies the 2 pF physical capacitor into a 100 pF effective capacitor at the input, which will cause the gain to drop at much lower frequencies.